

Question 01

Find a customer's city with VLOOKUP

Customer data is often stored in a reference table that you query by ID. VLOOKUP is a quick way to pull information from that table without searching row by row.

VLOOKUP searches for a value in the first column of a table and returns a value from another column in the same row. Its syntax is:

```
=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])
```

- **Lookup_value:** the value you want to find (the customer ID)
- **table_array:** the range that contains the data to search
- **col_index_num:** which column of the table to return (counting from the left edge of table_array)
- **range_lookup:** use FALSE for an exact match

Your task

The customer directory is in cells **A1:D7**. Below it, cell **B9** contains a customer ID, and cell **B10** is where you need to enter your formula.

1. Click cell **B10**.
2. Write a VLOOKUP formula that looks up the customer ID in **B9** in the directory table.
3. Return the city for that customer.
4. Use **FALSE** as the last argument so Excel only matches the exact customer ID.

Customer ID	Customer name	City	State
C1001	Riverside Bakery	Portland	OR
C1002	Summit Tech	Seattle	WA
C1003	Harbor Supplies	Boston	MA
C1004	Prairie Foods	Denver	CO
C1005	Coastal Gifts	Miami	FL
C1006	Maple Imports	Chicago	IL

Customer ID	C1004
City	Answer Here

ANSWER: =VLOOKUP(B9,A2:D7,3,FALSE())

QUESTION 2

Look up a product with XLOOKUP

Beginner

When you need to find a value in a table, XLOOKUP is Excel's most modern lookup tool. It has a cleaner syntax than VLOOKUP and works in any direction.

XLOOKUP searches for a value in one column and returns a value from a different column in the same row.

The three required arguments are:

=XLOOKUP(lookup_value, lookup_array, return_array)

- **lookup_value:** the value you want to find (a product code, a name, an ID, etc.)
- **lookup_array:** the column to search in
- **return_array:** the column to return a value from

Your task

A product catalog is listed in rows 1–8 (product code, product name, category). Below it, cell **B10** contains a product code you want to look up.

1. Click cell **B11**.
2. Write an XLOOKUP formula that finds the product code in **B10** within the product code column (**A2:A8**).
3. Return the matching value from the category column (**C2:C8**).
4. Press **Enter**.

When you're done, **B11** should show the category for that product code.

Product code	Product name	Category
PT-102	Push pins	Office supplies
MK-205	Dry-erase markers	Office supplies
WB-401	Dry-erase whiteboard	Office equipment

SC-108	Scissors	Office supplies
FD-312	File folders	Office supplies
ST-220	Stapler	Office equipment
NB-150	Spiral notebook	Office supplies

Look up this product code

WB-401

Category

Answer Here

ANSWER: =XLOOKUP(B10,A2:A8,C2:C8)

QUESTION 03

XLOOKUP with exact match

When managing inventory, you often have two separate lists: a master inventory table and a shorter reorder list. XLOOKUP makes it easy to pull values from one list into the other.

XLOOKUP searches for a value in a lookup column and returns a value from a return column in the same row. The three required arguments are:

=XLOOKUP(lookup_value, lookup_array, return_array)

- **lookup_value:** the value you want to find (a SKU, a name, an ID, etc.)
- **lookup_array:** the column to search in
- **return_array:** the column to return a value from

XLOOKUP uses exact match by default, so no extra argument is needed for that.

Your task

A warehouse inventory table is in rows 1–8 (SKU, Item name, Qty in stock). Below it, rows 10–14 show a reorder list with the same columns. The SKU and Item name columns are already filled in.

1. Click cell **C11**.
2. Write an XLOOKUP formula that finds the SKU from **A11** in the inventory SKU column (**\$A\$2:\$A\$8**).
3. Return the matching value from the Qty in stock column (**\$C\$2:\$C\$8**).
4. Copy the formula down through **C14**.

When you're done, cells **C11:C14** should show the current stock quantity for each item on the reorder list.

SKU	Item name	Qty in stock
SKU-1140	Cable ties (100-pack)	342
SKU-1141	Zip-lock bags (50-pack)	128
SKU-1142	Stretch wrap roll	56
SKU-1143	Packing tape (36-pack)	89
SKU-1144	Bubble wrap (25m)	203
SKU-1145	Label printer paper (500 sheets)	74
SKU-1146	Box cutter	165

SKU	Item name	Stock qty
SKU-1143	Packing tape (36-pack)	89
SKU-1140	Cable ties (100-pack)	342
SKU-1146	Box cutter	165
SKU-1144	Bubble wrap (25m)	203

ANSWER: =XLOOKUP(A11,\$A\$2:\$A\$8,\$C\$2:\$C\$8)

QUESTION 04

XLOOKUP left lookup

Intermediate

Sometimes the value you want is to the left of the value you have. VLOOKUP cannot return left, but XLOOKUP can.

XLOOKUP lets you pick the lookup column and the return column separately:

=XLOOKUP(lookup_value, lookup_array, return_array, "if_not_found")

Your task

You have a product table in **A2:D5**. The product name is in column **B**, but the SKU you want is in column **A**.

1. In cell **B8**, return the SKU for the product in **B7**.

2. If there is no match, return **Not found**.

Tip: Lock the table ranges with **\$** so the formula stays stable as you edit.

SKU	Product	Category	Price
SK-100	Travel Mug	Drinkware	12
SK-200	Desk Mat	Office	18
SK-300	USB Hub	Electronics	24
SK-400	Laptop Stand	Office	35

Product to find:

Desk Mat

SKU result:

ANSWER: `=XLOOKUP(B7,B2:B5,A2:A5)`

QUESTION 05

XLOOKUP with default

Intermediate

In inventory datasets, you'll often get a list of SKUs that includes old or discontinued items. Instead of returning an error, you can make your lookup return a clean default value.

XLOOKUP can return a fallback result using its optional **if_not_found** argument:

`=XLOOKUP(lookup_value, lookup_array, return_array, "if_not_found")`

Your task

You have an item master list in columns **A:D** (SKU, Product, Category, Unit cost). On the right, there's a restock list of SKUs in **F2:F8**.

1. In cell **G2**, use **XLOOKUP** to return the **Product** name for the SKU in **F2**.
2. If the SKU isn't in the item master list, return **Discontinued** (not an error).
3. Fill the formula down through **G8**.

Tip: Lock the lookup ranges (with **\$**) so the formula copies down cleanly.

Final Answer Table

NW-1001	Box Cutter	Tools	6.25	NW-1001	Box Cutter
NW-1002	Packing Tape (48mm)	Packaging	3.4	NW-1008	Discontinued

NW-1003	Bubble Wrap Roll	Packaging	19.99	NW-1003	Bubble Wrap Roll
NW-1004	Pallet Wrap	Packaging	14.5	NW-1012	Discontinued
NW-1005	Safety Gloves (L)	Safety	8.75	NW-1005	Safety Gloves (L)
NW-1006	Label Printer Paper	Office	27.5	NW-9999	Discontinued
NW-1007	Warehouse Marking Tape	Safety	12	NW-1007	Warehouse Marking Tape

Answer: =XLOOKUP(F2,\$A\$2:\$A\$8,\$B\$2:\$B\$8,"Discontinued")

QUESTION 06

XLOOKUP with wildcard search

Intermediate

In inventory work, product names are not always entered consistently. Sometimes you only know part of a product name, but you still need to find the matching item quickly.

XLOOKUP can handle wildcard searches when you set the **match_mode** argument to **2**. The asterisk (*) wildcard means "any sequence of characters", so a partial term can match a longer product name that contains that text.

The full XLOOKUP syntax is:

=XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found], [match_mode])

For a wildcard search, you need to build a lookup value that includes wildcard characters around the search term, then use **2** for the match mode.

Your task

A product catalog is listed in rows 1 through 9. The partial search term is already entered in **B11**.

1. In **B13**, use **XLOOKUP** to return the product name in the catalog that contains the search term.
2. In **B14**, use **XLOOKUP** again to return the unit price for that same matched product.
3. In both formulas, build the lookup value yourself by adding an asterisk before and after the search term from **B11**.
4. In both formulas, search within the product name column and use wildcard matching with **match_mode** set to **2**.

5. When you're done, **B13** should show the matching product name and **B14** should show its unit price.

Product code	Product name	Category	Unit price
PK-101	Pallet Wrap Film	Packaging	22.4
CL-204	Industrial Cleaner Pro	Cleaning supplies	18.5
LB-220	Storage Bin Label Set	Storage	7.2
CL-318	Heavy-Duty Cleaner Spray	Cleaning supplies	9.75
SF-402	Safety Floor Sign	Safety	31
CL-509	Citrus Surface Cleaner	Cleaning supplies	12.6
TB-615	Tool Cart Liner	Maintenance	14.25
CW-730	Cleaning Wipes Bulk Pack	Cleaning supplies	24.9

Search term	Surface
-------------	---------

Matching product	
Unit price	

Answer :

Matching product	=XLOOKUP("'"&B11&"'",\$B\$2:\$B\$9,\$B\$2:\$B\$9,,2)
Unit price	=XLOOKUP("'"&B11&"'",\$B\$2:\$B\$9,\$D\$2:\$D\$9,,2)

QUESTION 07

Lookup error handling

Beginner

Missing items are normal. The key is returning a clean message instead of #N/A.

You can wrap VLOOKUP with IFNA, or you can use XLOOKUP's built-in **if_not_found** argument.

Your task

You have a small inventory list in **A2:C4** and a lookup SKU in **B6**.

1. In cell **B7**, write a VLOOKUP that returns the product name. Wrap it with IFNA so missing items return **Not found**.
2. In cell **B8**, write an XLOOKUP that returns the same result using its **if_not_found** argument.

Tip: Use exact match for VLOOKUP (the 4th argument set to FALSE).

Need some help?

SKU	Product	Stock
NW-10	Packing Tape	120
NW-11	Shipping Boxes	50
NW-12	Label Rolls	0

Lookup SKU:

NW-99

VLOOKUP + IFNA:

XLOOKUP not_found:

ANSWERS :

VLOOKUP + IFNA: =IFNA(VLOOKUP(B6,A2:C4,2,FALSE()),"NOT FOUND")

XLOOKUP not_found: =XLOOKUP(B6,A2:A4,B2:B4,"NOTFOUND")

QUESTION 08

Look up a manager with INDEX+MATCH

Intermediate

A common lookup task is pulling the right manager name into a form or workflow step. The tricky part is that your lookup value isn't always in the first column.

In this sheet, the roster is set up with **Manager** in column A and **Employee** in column B. That layout is common in exports, but it means a left-to-right lookup (like basic VLOOKUP) won't fit the problem.

INDEX + MATCH solves this by:

- Using **MATCH** to find the row where an employee name appears
- Using **INDEX** to return the manager from the matching row

Your task

Fill in the missing manager names for the three pay change requests:

1. Enter an INDEX-MATCH formula in **C12** that looks up the manager for the employee in **B12**.
2. Copy the formula down through **C14**.

Tip: Use exact match in MATCH, and lock the roster ranges so your formula copies cleanly.

Manager	Employee	Team	Pay grade
Maria Gomez	Ava Patel	HR Ops	P2
Maria Gomez	Jordan Lee	HR Ops	P2
Thomas Nguyen	Elena Rossi	Payroll	P3
Thomas Nguyen	Noah Kim	Payroll	P3
Priya Shah	Liam Chen	Benefits	P2
Priya Shah	Sophia Martinez	Benefits	P2
David Brown	Mia Johnson	Recruiting	P1
David Brown	Ethan Wilson	Recruiting	P1

Pay change requests		
Employee	Manager (fill)	Reason
Noah Kim		Missing manager in request form
Mia Johnson		New manager assignment needed
Jordan Lee		Approval workflow requires manager

Answers: INDEX + MATCH: =INDEX(\$A\$2:\$A\$9,MATCH(A13,\$B\$2:\$B\$9,0))

SAME WE CAN DO WITH XLOOK UP

QUESTION 08

INDEX-MATCH left lookup

Intermediate

It's a common sight: exports where the columns aren't in the order you wish they were. For example, email appears to the left of customer ID, so a basic VLOOKUP can't return it.

INDEX + MATCH solves this cleanly:

- **MATCH** finds the row number where a customer ID appears
- **INDEX** returns the email from that same row

Your task

You have a customer directory in **A1:D9**, where **Email** is in column A and **Customer ID** is in column B.

Fill the missing emails for the campaign tasks:

1. In **C12**, write an INDEX-MATCH formula that returns the email for the customer ID in **B12**.
2. Copy the formula down through **C14**.

Use exact match in MATCH, and lock the directory ranges so your formula copies correctly.

Email	Customer ID	Customer name	Segment
alex.garcia@northwind.com	C-1042	Alex Garcia	SMB
bianca.chen@northwind.com	C-1047	Bianca Chen	SMB
diego.souza@northwind.com	C-1050	Diego Souza	Mid-market
emma.johansson@northwind.com	C-1055	Emma Johansson	Enterprise
farah.khan@northwind.com	C-1059	Farah Khan	Mid-market
henry.patel@northwind.com	C-1061	Henry Patel	SMB
ingrid.larsen@northwind.com	C-1068	Ingrid Larsen	Enterprise
jordan.reed@northwind.com	C-1072	Jordan Reed	Mid-market

Campaign task	Customer ID	Email (fill)	Note
Renewal reminder	C-1068		Email needed for send list

Upsell outreach	C-1047		Confirm contact before sending
Onboarding check-in	C-1050		Customer asked for follow-up

ANSWERS: =INDEX(\$A\$2:\$A\$9,MATCH(B12,\$B\$2:\$B\$9,0))

QUESTION 09

INDEX-MATCH with MATCH in both directions

Intermediate

Shipping teams often keep delivery prices in a matrix, with one category listed by row and another by column. A two-way lookup pulls the value where those two labels intersect, which makes this pattern useful for rate cards, schedules, and pricing tables.

In this exercise, you'll use **INDEX** with two **MATCH** functions to return the correct shipping rate. One **MATCH** should find the row for the selected zone, and the other should find the column for the selected weight bracket.

A common pattern looks like this:

=INDEX(data_range, MATCH(row_value, row_labels, 0), MATCH(column_value, column_labels, 0))

Build your formula in **B10** using the lookup inputs already provided in the query area. Return the shipping rate at the intersection of the chosen zone and weight bracket.

Want to see both **MATCH** results before **INDEX** uses them? [Follow INDEX MATCH with multiple criteria step by step.](#)

Zone	0-5 kg	5-10 kg	10-20 kg	20+ kg
Local	5	7	10	14
Regional	9	12	16	22
National	14	19	28	40
International	24	33	48	68
Express	30	42	60	85

Lookup zone National

Lookup weight 10-20 kg

Shipping rate

ANSWER:

- **USING INDEX MATCH BOTH DIRECTION:**
`=INDEX(B2:E6,MATCH(B8,A2:A6,0),MATCH(B9,B1:E1,0))`
- **USING XLOOKUP :** `=XLOOKUP(B8,A2:A6,D2:D6)`

QUESTION 10

XLOOKUP multiple criteria

Advanced

Sales Ops

In real-world scenarios, you often need to look up data based on multiple criteria. While XLOOKUP is typically used for single-criterion lookups, you can nest XLOOKUP functions to handle two-dimensional lookups.

The key insight is that XLOOKUP's `return_array` parameter can itself be an XLOOKUP result. This creates a powerful two-way lookup:

`=XLOOKUP(row_lookup, row_range, XLOOKUP(col_lookup, col_headers, data_range))`

The inner XLOOKUP finds the correct column, returning an entire column of data. The outer XLOOKUP then finds the correct row within that column.

Your task

You have a pricing table with products in column A and quarterly prices in columns B through E. The goal is to build a dynamic lookup that returns the price for any product in any quarter.

In cell **H2**, enter the product name "Widget D".

In cell **H3**, enter the quarter "Q3".

In cell **H5**, write a nested XLOOKUP formula that:

1. Uses the product in H2 to find the correct row
2. Uses the quarter in H3 to find the correct column
3. Returns the price at the intersection

The formula should work for any product and quarter combination.

Product	Q1	Q2	Q3	Q4	Find price for:
Widget A	25.99	27.49	26.99	28.99	Product:
Widget B	34.5	35	36.25	37	Quarter:
Widget C	18.75	19.25	19	20.5	
Widget D	42	43.5	44	45.25	Price:
Widget E	56.99	58.49	59.99	61.5	
Widget F	12.25	12.75	13	13.5	
Widget G	89.99	92.5	94	96.75	

ANSWER : `XLOOKUP(H2,A2:A8,XLOOKUP(H3,B1:E1,B2:E8))`

QUESTION 11

Return multiple columns

Advanced

In analytical datasets, you rarely look up a single field. More often you need to pull a small “record” (multiple columns) for an ID, and you want it to stay maintainable as the dataset grows.

Your sheet contains a customer table in **A9:E15**. The lookup key is **Customer ID** (column A). The fields you need to return are **Name, Email, Phone** (columns B–D).

Your task

Write a single formula in **C5** that:

- Uses the Customer ID in **B3** as the lookup value
- Returns **Name, Email, and Phone** as a horizontal array that spills across **C5:E5**
- Pulls the values from the customer table (do not hardcode any customer details)

Your solution should keep working if the value in **B3** changes to any other valid ID in the table.

Customer Lookup

Look up a customer ID and return Name, Email, and Phone in one spilled row.

Customer ID	1004			
		Name	Email	Phone

Customer Table				
Customer ID	Name	Email	Phone	Segment
1001	Ava Patel	ava.patel@northwind.com	(555) 010-0421	Standard
1002	Noah Kim	noah.kim@northwind.com	(555) 010-0673	Standard
1003	Sofia Martinez	sofia.martinez@northwind.com	(555) 010-0905	Premium
1004	Keira Lewis	keira.lewis@northwind.com	(555) 010-1188	Premium
1005	Liam Chen	liam.chen@northwind.com	(555) 010-1430	Standard
1006	Maya Singh	maya.singh@northwind.com	(555) 010-1762	Standard
1007	Ethan Brooks	ethan.brooks@northwind.com	(555) 010-2014	Enterprise

ANSWER : `=XLOOKUP(B3,A9:A15,B9:D15)`

QUESTION 12

VLOOKUP for price

Beginner

When working with Excel, you'll often need to look up a value in a table. In this example, we will pull a price from a product catalog using a product code. VLOOKUP is a quick way to do that.

VLOOKUP searches for a value in the *first column* of a table, then returns a value from another column in the same row.

The syntax of VLOOKUP looks like this:

`=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])`

Your task

Fill in the unit price for the order line:

1. Look at the product catalog in cells **A2:C6**.
2. Click cell **B9** (Unit price).
3. Use **VLOOKUP** to find the price for the product code in **A9**.
4. Use the catalog range **A2:C6** as your table.
5. Return the value from the **3rd column** of that table (the Unit price column).
6. Use an **exact match** lookup (so the code must match exactly).

Product code	Product	Unit price (\$)
P-1001	Notebook	\$3.50
P-1002	Pen	\$1.25
P-1003	Marker	\$2.00
P-1004	Stapler	\$7.99
P-1005	Tape dispenser	\$4.60
Order lookup		
Product code	Unit price (\$)	
P-1004	\$7.99	

ANSWER: =VLOOKUP(A5,A1:C6,3,1)

QUESTION 13

Use VLOOKUP to find an employee's department

Beginner

Many organizations keep a simple employee directory (ID to department). Instead of searching manually, you can use VLOOKUP to pull the right department automatically.

VLOOKUP syntax:

=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])

In this sheet:

- The **Directory** table is in **A2:B7** (headers in row 2, data in rows 3 to 7)
- The **Lookup** table is in **A10:B14**

Your task

1. Click cell **B11**.
2. Type a **VLOOKUP** formula that uses the employee ID in **A11** and searches the directory in **A3:B7**.
3. Return the **Department** from the directory (the second column of the directory table).
4. Use **exact match** so Excel only matches the exact employee ID.
5. Lock the directory range with **\$** so you can copy the formula.
6. Copy your formula down from **B11** to **B14**.

When you're done, cells **B11:B14** should show the correct departments for each employee ID.

Directory

Employee ID	Department
1040	Finance
1042	HR
1051	IT
1063	Operations
1038	Payroll

Lookup

Employee ID	Department (fill this)
1042	HR
1038	Payroll
1051	IT
1040	Finance

ANSWER ARE IN GREEN FIELD : **=VLOOKUP(A11,\$A\$3:\$B\$7,2,FALSE)**

QUESTION 14

HLOOKUP for quarterly targets

Beginner

VLOOKUP searches down a column to find a match. **HLOOKUP** does the same thing, but searches across a row instead. It's the right tool when your reference data is laid out in rows rather than columns.

HLOOKUP syntax:

=HLOOKUP(lookup_value, table_array, row_index_num, [range_lookup])

- **lookup_value**: the quarter name to search for
- **table_array**: the range that contains both the header row and the values row
- **row_index_num**: which row within the table to return a value from (1 = first row, 2 = second row)
- **range_lookup**: use **FALSE** for an exact match

In this sheet, the targets table is in **A1:E2**. Row 1 has a row-label column in A1 and the quarter headers Q1 through Q4 in columns B through E. Row 2 contains the corresponding sales targets. The quarter you want to look up is in **B4**.

Your task

1. Click cell **B5**.
2. Write an **HLOOKUP** formula that looks up the quarter in **B4** within the targets table in **A1:E2**.
3. Return the sales target from row 2 of the table.
4. Use **FALSE** for an exact match.

When you're done, cell **B5** should show the sales target for the quarter in B4.

Metric	Q1	Q2	Q3	Q4
Sales target	85000	95000	110000	120000

Quarter Q3

Target 110000

ANSWER: =HLOOKUP(B4,A1:E2,2,FALSE)

QUESTION 15 VLOOKUP with approximate match

Intermediate

Shipping teams often price delivery in tiers (for example: 0–1 lb, 1–5 lb, 5–10 lb, and so on). In Excel, this is a classic use case for VLOOKUP with an approximate match.

In this sheet, you have:

- A list of orders with package weights in **C5:C9**
- A tiered rate table in **G5:H9** (minimum weight → rate per lb)

Your task

Fill in **D5:D9** with a formula that looks up the correct **Rate (\$/lb)** based on each row's **Weight (lb)**.

Requirements:

1. Use **VLOOKUP** with **approximate match (TRUE)** so each weight picks the highest tier that is less than or equal to the weight.
2. Lock the rate table range so you can copy the formula down.
3. Copy your formula from **D5** down through **D9**.

When the rates are correct, the **Shipping cost** column will calculate automatically.

ANSWER :

=VLOOKUP(C5,\$G\$5:\$H\$9,2)

Orders					Rate table	
Order #	Zone	Weight (lb)	Rate (\$/lb)	Shipping cost	Min weight (lb)	Rate (\$/lb)
SO-10421	Domestic	0.8	6.5	5.2	0	6.5
SO-10422	Domestic	2.4	5.75	13.8	1	5.75

SO-10423	Canada	6.2	5	31	5	5
SO-10424	EU	12	4.25	51	10	4.25
SO-10425	Domestic	24.5	3.75	91.875	20	3.75